

1 Chapter 1

Nick's Question 1.1 Proof of lemma 1.8.4

The proof is very nice.

2 Chapter 2

Nick's Question 2.1 In the proof of Theorem 2.4.3 on the bottom of page 34 there are some claims made that I am unable to verify why they are true. First, we are considering

$$X_t = X_0 + \int_0^t u(s)dB_s$$

and here $u(s)$ only satisfies

$$\mathbb{P}\left(\int_0^t u(s)^2 ds < \infty\right) = 1$$

so, as in the Kuo text, this stochastic integral exists and a limit in probability of a sequence of stochastic integrals $\int_0^t u_n(s)dB_s$ where the u_n are simple and are in L^2_{ad} and hence $\int_0^t u_n(s)dB_s$ is a martingale and we know the first and second moment. But for the limit $\int_0^t u(s)dB_s$ we know nothing about the moments. But in this text, it is claimed that

$$\mathbb{E}\left[\int_{t_i}^{t_{i+1}} u^2(s)ds - \left(\int_{t_i}^{t_{i+1}} u(s)dB_s\right)^2 \middle| \mathcal{F}_i\right] = 0$$

and I have never seen this claimed in any other text.

Note that u is bounded (we are assuming localization has been done). Hence, all moments exists.

Nick's Question 2.2 In the proof of theorem 2.5.1, why is it true that

$$E\left(\int_0^t |u'_n(B_s) - 1_{(x,\infty)}(B_s)|^2 ds\right) \leq \int_0^t \mathbb{P}\left(|B_s - x| \leq \frac{2}{n}\right) ds$$

Please explore the mollifier f and f_n . Plot these u'_n and see how could it differ from $1_{(x,\infty)}$.

Nick's Question 2.3 What is the motivation on page 41 behind the following definition of the backward ito integral

$$\int_0^T u(s)d\hat{B}_s = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-2} \left(\frac{n}{T} \int_{(j+1)T/n}^{(j+2)T/n} u(s)ds \right) (B_{(j+1)T/n} - B_{jT/n})$$

The integral in the parenthesis is the average of u_s on $((j+1)T/n, (j+2)T/n)$.

Let's skip this part. I know nothing on the backward SDE and not particular interested either. It has applications in finance. If you don't do finance, I don't see the point here.

Nick's Question 2.4 (Problem 2.7)

This problem follows easily from Levy's Characterization of Brownian Motion. But I can't figure out how to do this problem with the hint given.

This is a nice problem. The hints is to ask you to find the distribution of $X_t - X_s$ through the characteristic function. Once apply Ito's formula, take expectation and remove the martingale part, you should arrive at an integral equation. Take derivatives, you arrive at a differential equation. The solution for $a_s(t) := \mathbb{E}(e^{i\lambda(X_t - X_s)})$ is

$$a_s(t) = e^{-\frac{\lambda^2}{2}(t-s)}.$$

Nick's Question 2.5 (Problem 2.15)

First we consider the 2 - d Ito process

$$\begin{pmatrix} M_s \\ N_s \end{pmatrix} = \begin{pmatrix} E(X^p) \\ E(Y^{q'}) \end{pmatrix} + \int_0^t \begin{pmatrix} \varphi_s & 0 \\ 0 & \psi_s \end{pmatrix} \begin{pmatrix} d\hat{B}_s \\ dB_s \end{pmatrix}$$

and then following the hint we apply Ito's formula to $f(x, y) = x^{1/p}y^{1/q'}$. We get the following derivatives:

- $f_x = \frac{1}{p}x^{\frac{1}{p}-1}y^{\frac{1}{q'}}$
- $f_{xx} = \frac{1}{p}\left(\frac{1}{p}-1\right)x^{\frac{1}{p}-2}y^{\frac{1}{q'}}$
- $f_y = \frac{1}{q'}x^{\frac{1}{p}}y^{\frac{1}{q'}-1}$
- $f_{yy} = \frac{1}{q'}\left(\frac{1}{q'}-1\right)x^{\frac{1}{p}}y^{\frac{1}{q'}-2}$

Now putting this into Ito's Formula gives us

$$\begin{aligned} M_t^{1/p} N_t^{1/q'} &= \|X\|_p \|Y\|_{q'} + \frac{1}{p} \int_0^t M_s^{\frac{1}{p}-1} N_s^{\frac{1}{q'}} \varphi_s d\hat{B}_s + \frac{1}{q'} \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-1} \psi_s dB_s \\ &\quad + \frac{1}{2} \frac{1}{p} \left(\frac{1}{p}-1\right) \int_0^t M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}} \varphi_s^2 ds + \frac{1}{2} \frac{1}{q'} \left(\frac{1}{q'}-1\right) \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-2} \psi_s^2 ds \end{aligned}$$

I am not really sure where to continue from here. I see that if we let $t \rightarrow \infty$ then $M_t^{1/p} N_t^{1/q'} \rightarrow XY$ so if we took the expected value of both sides then we would partially have what we are looking for since $E(XY)$ is on the left and $\|X\|_p \|Y\|_{q'}$ is on the right. But I am not sure about the other terms. Im guessing the Ito integrals in the right have 0 expectation but I am not sure if that is in fact true since I dont see how it is easily seen that the integrands are in $L^2(\mathcal{P})$.

What I did next was see what $\hat{B}_s$ was equal too and for this we define another Ito Process

$$\begin{pmatrix} B_s \\ \tilde{B}_s \end{pmatrix} = \int_0^t \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} dB_s \\ d\tilde{B}_s \end{pmatrix}$$

and applied Itos formula to $f(t, x, y) = e^{-t}x + \sqrt{1 - e^{-2t}}y$ which gives us

$$\hat{B}_s = e^{-t}B_s + \sqrt{1 - e^{-2t}}\tilde{B}_s = \int_0^t \frac{e^{-2s}}{\sqrt{1 - e^{-2s}}}\tilde{B}_s - e^{-s}B_s ds + \int_0^t e^{-s}dB_s + \int_0^t \sqrt{1 - e^{-2s}}d\tilde{B}_s$$

However I am not sure if this step is necessary.

The followings are some of my thoughts: First of all, we need the absolute values:

$$\begin{pmatrix} M_s \\ N_s \end{pmatrix} = \begin{pmatrix} \mathbb{E}(|X|^p) \\ \mathbb{E}(|Y|^{q'}) \end{pmatrix} + \int_0^t \begin{pmatrix} \varphi_s & 0 \\ 0 & \psi_s \end{pmatrix} \begin{pmatrix} d\hat{B}_s \\ dB_s \end{pmatrix}.$$

Second, you forget

$$f_{xy} = \frac{1}{p} \frac{1}{q'} x^{-1/(p')} y^{-1/q}.$$

Hence, by Ito's formula, we will have

$$\begin{aligned} M_t^{1/p} N_t^{1/q'} &= \|X\|_p \|Y\|_{q'} + \frac{1}{p} \int_0^t M_s^{\frac{1}{p}-1} N_s^{\frac{1}{q'}} \varphi_s d\hat{B}_s + \frac{1}{q'} \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-1} \psi_s dB_s \\ &\quad + \frac{1}{2} \frac{1}{p} \left(\frac{1}{p} - 1 \right) \int_0^t M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}} \varphi_s^2 ds + \frac{1}{2} \frac{1}{q'} \left(\frac{1}{q'} - 1 \right) \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-2} \psi_s^2 ds \\ &\quad + \frac{1}{pq'} \int_0^t M_s^{-\frac{1}{p'}} N_s^{-\frac{1}{q}} \psi_s \phi_s d\langle B, \hat{B} \rangle_s \\ &=: \|X\|_p \|Y\|_{q'} + \sum_{i=1}^5 I_i. \end{aligned}$$

Upon taking expectation on both sides, we see that

$$LHS = \mathbb{E}(|X_t Y_t|) \geq \mathbb{E}(X_t Y_t).$$

We only need to prove that

$$\mathbb{E} \left(\sum_{i=1}^5 I_i \right) \leq 0,$$

which is clear for $i = 1, 2$, in which case we have the expectation equal to zero. Now for the rest three parts, we see that

$$I_3 + I_4 + I_5 = -\frac{1}{2pq'} M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}-2} \left(N_s^2 \varphi_s^2 \frac{q'}{p'} - 2M_s N_s \phi_s \psi_s e^{-s} + M_s \psi_s^2 \frac{p}{q} \right),$$

where we have used the fact that $d\langle B, \hat{B} \rangle_s = e^{-s} ds$. Hence, the condition become the condition that the quadratic form has no solution:

$$\Delta = 4e^{-2s} - 4\frac{q'p}{p'q} = 4 \left(e^{-2s} - \frac{p-1}{q-1} \right) \leq 0$$

which is equivalent to

$$q \leq e^{2s}(p-1) + 1.$$

Hence, the largest q that makes the inequality work is $q = e^{2s}(p-1) + 1$.

3 Chapter 3

Nick's Question 3.1 (Smooth Cylindrical Random Variable) Following chapter 3 of the Kuo text, denote C the Wiener space of continuous functions from $[0,1]$ to $\mathbb{R}$, $\mathcal{R}$ the collection cylindrical sets

$$A = \{\omega \in C : (\omega(t_1), \dots, \omega(t_n)) \in U \mid t_i \in [0, 1], n \in \mathbb{N}, U \in \mathcal{B}(\mathbb{R}^n)\}$$

and μ the associated Wiener measure. Consider the coordinate processes $B(t, \omega) = \omega(t)$ which are Brownian motions under $(C, \mathcal{B}(\mathcal{R}), \mu)$. There is a theorem that can be found [here](#).

Theorem : The random variables

$$\{f(B(t_1, \cdot), \dots, B(t_n, \cdot)) \mid n \in \mathbb{N}, f \in \mathcal{S}(\mathbb{R}^n)\}$$

are dense in $L^2(C, \mu)$ where $\mathcal{S}(\mathbb{R}^n)$ are the Schwartz functions.

How does this relate to $\mathcal{S}$ defined on page 51 of the intro to Malliavin textbook? How are the cylindrical random variables defined in this case? Why is $\mathcal{S}$ dense in $L^2(\Omega)$? Even in Diffusions Markov Processes and Martingales - Roger and Williams, they define the cylindrical sigma field in the same way Kuo does- through the use of coordinate processes.

Nick's Question 3.2 Why is $(D_t F)_{t \geq 0}$ a stochastic processes in $L^2(\Omega; H)$ where $H = L^2(\mathbb{R}^+)$? $D_t F$ is defined as

$$D_t F = \sum_{i=1}^n \frac{\partial f}{\partial x} (B(h_1), \dots, B(h_n)) h_i(t)$$

where $h_i \in H$ and $B(h_i) = \int_0^\infty h_i dB_s$ is the standard Wiener Integral. So isnt $D_t F$ real valued since we are evaluating the $h_i(\cdot)$ at t ? Then shouldn't the $(D_t F)_{t \geq 0}$ be a $L^2(\Omega; \mathbb{R})$ valued stochastic process?

Nick's Question 3.3 (Proposition 3.2.4) $D_h(F) = \langle DF, h \rangle$ is defined only for $F \in \mathcal{S}$ since the D operator is only defined on $\mathcal{S}$. However in (3.5) in this proposition, we want to calculate $D_h(u)$ but $u \in \mathcal{S}_H$ and in this case $D_h(u)$ has not been defined. I am not sure how they are getting equality 2 and 3 from the first three centered equations on page 53.

Also how do you get the RHS of the last inequality in the second set of centered equalities on page 53.

Also as the bottom of page 52, we is it that

$$E \left(\sum_{i,j=1}^{\infty} D_{e_i} \langle u, e_j \rangle D_{e_j} \langle v, e_i \rangle \right) = E \left(\int_0^\infty \int_0^\infty D_s u_t D_t v_s ds dt \right)$$

What does $D_s u_t$ and $D_t v_s$ mean?

Partial Proof of

$$D_h(\delta(u)) = \langle u, h \rangle + \delta(D_h u).$$

Since $u = \sum F_j h_j$ we just apply the definition of δ to get

$$D_h(\delta(u)) = D_h \left(\sum_{j=1}^n F_j B(h_j) - \sum_{j=1}^n \langle DF_j, h_j \rangle \right)$$

$$\begin{aligned}
&= \sum_{j=1}^n D_h(F_j B(h_j)) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \\
&= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle
\end{aligned}$$

where the third inequality follows from the product rule and $D_h(B(h_j)) = \langle h, h_j \rangle$. Now we focus on the last term from the last line above.

$$\sum_{j=1}^n D_h \langle DF_j, h_j \rangle = \sum_{j=1}^n \langle D. \langle DF_j, h_j \rangle, h(\cdot) \rangle$$

Before we continue, we simplify $D. \langle DF_j, h_j \rangle$. We may assume that $F_j = f^j(B(h_1^j(t), \dots, B(h_n^j(t))))$ where $f^j \in C_p^\infty(\mathbb{R}^n)$ and $h_i^j \in H$.

$$\begin{aligned}
D. \langle DF_j, h_j \rangle &= D. \left\langle \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(\circ), h_j(\circ) \right\rangle \\
&= \sum_{i=1}^n \langle h_i^j(\circ), h_j(\circ) \rangle D. \left(\frac{\partial f^j}{\partial x_i} \right)
\end{aligned}$$

We now plug this in above to get

$$\begin{aligned}
\sum_{j=1}^n \langle D. \langle DF_j, h_j \rangle, h(\cdot) \rangle &= \sum_{j=1}^n \sum_{i=1}^n \langle h_i^j(\circ), h_j(\circ) \rangle \langle D. \left(\frac{\partial f^j}{\partial x_i} \right), h(\cdot) \rangle \\
&= \sum_{j=1}^n \sum_{i=1}^n \left\langle h_i^j(\circ) \langle D. \left(\frac{\partial f^j}{\partial x_i} \right), h(\cdot) \rangle, h_j(\circ) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D. \left(\sum_{i=1}^n h_i^j(\circ) \frac{\partial f^j}{\partial x_i} \right), h(\cdot) \right\rangle, h_j(\circ) \right\rangle \\
&= \sum_{j=1}^n \left\langle \langle D. (DF_j), h(\cdot) \rangle, h_j(\circ) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_h (DF_j), h_j(\circ) \right\rangle
\end{aligned}$$

and if I haven't made a mistake, then what we have shown so far is that

$$\begin{aligned}
D_h(\delta(u)) &= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \\
&= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n \left\langle D_h (DF_j), h_j \right\rangle
\end{aligned}$$

Also, the above is exactly line two of the first centered equations on page 53.

Now we need to show that the last two terms add up to $\delta(D_h(u))$. This is not defined in the text but I think that we need to define $D_h u$ as the following

$$D_h u = D_h \sum_{j=1}^n F_j h_j(t) = \sum_{j=1}^n h_j(t) D_h(F_j)$$

Thus now $D_h(u)$ is of the form $\sum F_j h_j$ and so we may apply the definition of δ to it to get the following

$$\delta(D_h u) = \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n \langle D_\circ(D_h F_j), h_j(\circ) \rangle$$

Now we focus in the last term of the right hand side. From the above we note that we want to show that

$$\sum_{j=1}^n \langle D_\circ(D_h F_j), h_j(\circ) \rangle = \sum_{j=1}^n \left\langle D_h(D F_j), h_j \right\rangle$$

However I have not been able to how that this is true.

Nick's Question 3.4 (Why is δ linear?) On page 55 it says that δ is linear. First, for $u = \sum_{j=1}^n F_j h_j$ we have

$$\delta(u) = \sum_{j=1}^n F_j B(h_j) - \sum_{j=1}^n \langle D F_j, h_j \rangle_H$$

Now if we have a $v = \sum_{j=1}^n G_j h'_j$ how does $\delta(u + v) = \delta(u) + \delta(v)$. Also how could we write $\sum_{j=1}^n F_j h_j + \sum_{j=1}^n G_j h'_j = \sum_{j=1}^n K_j k_j$ where $K_j \in \mathcal{S}$ and $k_j \in H$

Do we do this through the relationship $E(\langle D F, u \rangle) = E(\delta(u) F)$ in the sense that

$$\begin{aligned} E(\delta(u + v) F) &= E(\langle D F, u + v \rangle) \\ &= E(\langle D F, u \rangle) + E(\langle D F, v \rangle) \\ &= E(\delta(u) F) + E(\delta(v) F) \\ &= E((\delta(u) + \delta(v)) F) \end{aligned}$$

and if this is true for every F then I believe we can say $\delta(u + v) = \delta(u) + \delta(v)$?